AHMED ALVEE

Solutions Architect | AI & Automation Engineering
Rotterdam, Netherlands | +31 6 4309 7073 | ahmedalv94@gmail.com
linkedin.com/in/alveeahmed | github.com/alvee1994 | alvee1994.github.io/portfolio

Summary

Solutions architect who built AWS data pipelines and LLM workflows for clinics, and took them live with customers on site. Led the first overseas deployment at Mesh Bio and a team of 2-3 solutions engineers. AWS Certified Solutions Architect – Associate. Completing an MBA at Rotterdam School of Management.

Skills

Cloud: AWS (Lambda, Step Functions, SQS, SNS, EventBridge), Terraform
AI and LLM: Model Context Protocol (MCP), Amazon Bedrock, LangGraph, LangChain, Gemini, Pydantic, ChromaDB, n8n, Claude Code, Cursor
Integration: REST APIs, Postman, HL7, MLLP, DICOM, Microsoft Graph API
Languages and tools: Python, C++, MongoDB, Git, Linux, Qt, ROS

Professional Experience

Mesh Bio Pte Ltd — Singapore, SG
Solutions Architect — 05/2022 – 12/2025
Singapore health-tech company serving private healthcare providers. Its health screening workflow scored patient risk and built screening reports.

- Cut per-submission time from about 1 hour to 6-10 minutes, including human review, for 30+ clinics by building an AI clean-up pipeline from scratch (LangGraph, Claude on Bedrock, Pydantic, ChromaDB) in which a human reviewer checked and submitted every file.
- Cut the time to onboard a new lab and clinic combination from about a week to a day by replacing a hardcoded Lambda per combination with a config-driven AWS pipeline (Lambda, SNS, SQS, EventBridge Pipes, Step Functions, HL7 integration) that handled about 10,000 records an hour and stayed inside Lambda and MongoDB connection limits.
- Kept wrong data out of patient profiles with record count checks and exception handling, and resolved a lab vendor's unannounced system switch in one day.
- Built the MLLP endpoint that receives HL7 from lab vendors and writes files to S3, and handled hourly scheduled files plus on-demand submissions for the lab, radiology and clinic pipelines.
- Turned the first overseas customer into a lighthouse account that opened follow-on regional projects by recommending a headless API-based workflow and leading 3 weeks of on-site deployment.
- Owned the integration APIs that became the distribution partner's primary way to integrate new clients; joined technical sales calls to map customer tech stacks and integration points.
- Hired, trained and supervised a team of 2-3 solutions engineers as they took over and expanded the APIs and pipelines.

Creative Medtech Solutions Pte Ltd — Singapore, SG
Software Engineer — 09/2021 – 04/2022
Singapore medtech company building a robotic system that ablates cancer cells with high intensity focused ultrasound (HIFU).

- Owned the annotation and user-interaction layer of the imaging system for a HIFU cancer-ablation robot: marking and measuring ablation targets, and DICOM export.

Singapore University of Technology and Design — Singapore, SG

Research Assistant — 02/2018 – 08/2021

University research lab building robotics and wearable systems, working with teams of research assistants and PhD students.

- Mapped 2 floors and 5 rooms in under 6 minutes with a 12+ robot system, owning its locomotion code.
- Recognized 7 hand gestures to command unmanned vehicles with an ML pipeline (Python, ROS, StretchSense).
- Contributed C++ and hardware engineering to 2 more robotics builds: a legged mapping robot and a custom PCB for 3D-knitted wearables.

MBA Consulting Project

RSM Living Management Project — Rotterdam, Netherlands

AI Strategy Consultant (Client: Truvalu) — 06/2026 – 07/2026

MBA consulting project for Truvalu, a Netherlands-based impact investor supporting agri-food SMEs in emerging markets.

- Projected about a 45% cut in monitoring time for an impact investor with a 5-stage AI monitoring workflow, designed as part of a 5-person MBA team.
- Engineered and demoed AI tools and agents (Gemini, Claude, Tally.so) spanning all 5 workflow stages, from data clean-up to reporting.
- Co-led the final client presentation to Truvalu leadership.

Education

Rotterdam School of Management, Erasmus University — Rotterdam, Netherlands

Master of Business Administration (MBA) — 01/2026 – 03/2027

Singapore University of Technology and Design — Singapore, SG

BEng (Honours), Engineering Product Development — 05/2014 – 08/2017

Certifications

- AWS Certified Solutions Architect – Associate, Amazon Web Services (10/2023).